Manas Tiwari

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in scalable backend design, full-stack MERN development, and agentic AI workflows. Proven ability to architect high-throughput, decoupled processing systems using **FastAPI**, **TaskIQ**, and **Redis** to efficiently manage heavy LLM workloads and distributed asynchronous pipelines.

EDUCATION

Indian Institute of Information Technology, Ranchi ***2023-2027***
B. Tech in Computer Science and Engineering CGPA: 9.35

St. Joseph's School, Greater Noida ***2008-2023***
10th and 12th (CISCE) Percentage: 95%

EXPERIENCE

Full-Time Intern - Automation Projects | BizeeBuy – [Intellifi Technologies] *(hybrid)* **April 2026 – July 2026**

- Engineered an intelligent database querying assistant using LangGraph and FastAPI to dynamically translate natural language queries into optimized, read-only MySQL syntax and data visualization upon execution.
- Architected a decoupled processing pipeline using **TaskIQ** and **Redis** to offload heavy LLM execution from the primary HTTP request-response loop, successfully scaling the system to support 2,000+ total users and **100+ concurrent query requests.** Integrated **distributed rate limiting** to protect API quotas.
- Built an interactive frontend using HTML, Bootstrap, and JS, integrating **ApexCharts** to dynamically render multi-axis visualizations.
- Implemented a robust security validation layer utilizing **sqlglot AST parsing** to neutralize prompt and SQL injection vulnerabilities, while enforcing strict **role-based access control (RBAC)** across isolated databases.

TECHNICAL SKILLS

Programming Languages: C++, TypeScript, JavaScript, Python.
Web Technologies: React.js, Next.js, Redux, REST API, Node.js, Express.js, WebSocket, WebRTC, JWT authentication, FastAPI.
AI & Frameworks: Retrieval-Augmented Generation, Generative AI, Agentic AI workflows, LangChain, LangGraph, ChromaDB, FAISS, Hugging Face, Llama 3, Gemini.
DevOps & Tools: Docker, TaskIQ, Postman.
Databases & Cloud: MongoDB (Mongoose), Cloudinary, SQL(MySQL), Redis, Vercel, Render.
Computer Science Core: DSA, DBMS, Operating Systems, Computer Networks, System Design.

PERSONAL PROJECTS

WatchNest - *A modern feature-rich video streaming platform*

Tech Stack: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, JWT, Redux, Cloudinary

- Engineered a high-performance video platform using the **MERN stack**, implementing **JWT-based authentication** with HTTP-only cookies and **bcrypt** hashing for industry-standard security.
- Configured **Cloudinary** for efficient media (photo/video) upload, storage, and retrieval with optimization support.
- Designed a scalable **Entity-Relationship (ER) model** using **MongoDB** and Mongoose to efficiently manage diverse user and media data. Optimized query execution and data retrieval speeds by implementing strategic **indexing** and complex **aggregation pipelines**.
- Developed a highly responsive UI featuring viewport observation, which significantly reduced redundant API calls and improved overall frontend performance.

Pulse Chat – *Real-Time Chat and Voice/Video Call application*

Tech Stack: HTML, CSS, Node.js, Express.js, Socket.IO and WebRTC

- Engineered a **real-time messaging platform** using Node.js and Express, featuring a custom user authentication system and a responsive frontend.
- Integrated **Socket.IO** to manage persistent bi-directional communication, supporting core features such as live online user tracking and dynamic unread message badges.
- **Implemented peer-to-peer voice and video calling** functionality by leveraging **WebRTC** protocols to facilitate high-quality, low-latency media streaming directly in the browser.

ACHIEVEMENTS

- Solved over **800+** problems across platforms like Leetcode, Codeforces and GeeksforGeeks.
- Attained (max) **1751 rating** on **Leetcode** and (max) **1230 rating** on **Codeforces**.
- Actively participated in the Smart India Hackathon (SIH) 2024, collaborating on innovative solutions to real-world challenges in a competitive national platform.